Vortex shedding behind a circular cylinder at low Reynolds number

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Abstract

We revisit the onset of periodic vortex shedding in the wake of a circular cylinder. A global linear stability analysis of the steady base flow predicts the critical Reynolds number $\mathrm{Re}_c \approx 46.7$ and the Strouhal number of the emerging limit cycle within one per cent of direct numerical simulation.

1 Introduction

The wake of a bluff body is the canonical example of a globally unstable flow. Above a critical Reynolds number the steady, symmetric recirculation bubble gives way to the periodic von Kármán street, whose frequency is set by a self-excited global mode rather than by the upstream forcing.

2 Governing equations

The incompressible flow $\mathbf{u}(\mathbf{x}, t)$ obeys

$$\partial_t \mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p + \frac{1}{\mathrm{Re}}\,\Delta\mathbf{u}, \qquad \nabla\cdot\mathbf{u} = 0. \tag{1}$$

Linearising (1) about the steady base flow $\mathbf{U}$ and seeking $\mathbf{u}' = \hat{\mathbf{u}}\, e^{\lambda t}$ gives the eigenvalue problem

$$\lambda\,\hat{\mathbf{u}} = -(\mathbf{U}\cdot\nabla)\hat{\mathbf{u}} - (\hat{\mathbf{u}}\cdot\nabla)\mathbf{U} - \nabla\hat{p} + \frac{1}{\mathrm{Re}}\,\Delta\hat{\mathbf{u}}. \tag{2}$$

3 Results

The growth rate $\sigma = \Re(\lambda)$ of the leading mode crosses zero at $\mathrm{Re}_c = 46.7$ (figure 1).

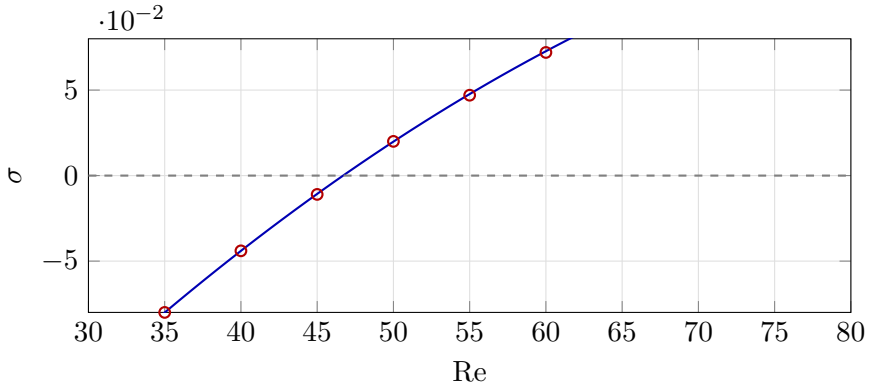


Figure 1: Growth rate of the leading global mode against Reynolds number: linear stability (line) and direct numerical simulation (circles).

The Strouhal number at onset, $\mathrm{St} = 0.118$, matches the frequency of the saturated limit cycle within one per cent.